

Electronic Companion for the Paper

Event-Triggered Holomorphic Embedding Safe Reinforcement Learning for OEF in Integrated Electric-Gas Systems

Documentation for Integrated Electric-Gas System Topology Benchmarks

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Prepared as supplementary documentation for the IEGS OEF reinforcement-learning benchmarks.

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1 Data Sources

This document describes the data sources of the three integrated energy system subsystems used in the current experiments. The electric-network topologies are taken from MATPOWER and ACTIVSg power-system data¹; the natural-gas network topologies are taken from the GasLib benchmark data².

2 Data Structure

2.1 Column Definitions for Gas-Network Data Tables

Table 1: Column Definitions of Gas-Network Data Tables.

Data table	Columns	Meaning and units
Well	0–6	gas-supply node, maximum/minimum gas production (MSm^3/h), cost coefficient, gas-source index, reference-well flag, and gas production
GenGas	0–3	unit type, gas-consumption efficiency, gas node, and electric-side bus; type 1 denotes a gas turbine and type 0 denotes a conventional generation plant
Demand	0–23	24-period gas-load multipliers; the initial values are all ones and are replaced by the prescribed operating profile
Load	0–5	gas node, maximum pressure, minimum pressure, base gas load, node-type flag, and reference-node pressure
Pipeline	0–3	pipeline inlet node, pipeline outlet node, length (m), and diameter (m); the pipeline tables below additionally report derived K_m and K_q coefficients computed from length and diameter
Comp	0–14	inlet node, outlet node, type, compression factor, α , k_1 , k_2 , compression-ratio limits, electric-power limits (only for electric-driven compressors), electric-side bus, loss, efficiency, and current ratio
P2G	0–7	electric-side load bus, gas node, efficiency, current power (MW), base capacity (MVA), upper power limit (MW), lower power limit (MW), and coupling-strength coefficient

3 Benchmark Overview

Table 2: Sizes of the Three IEGS Benchmarks.

Case	Power buses count	Power loads count	Gen. units	GT units	Gas nodes count	Gas loads count	Pipes branches	Comp units	P2G units
P14_G11	14	11	5	2	11	3	8	2	1
P57_G40	57	42	7	2	40	29	39	6	1
P200_G135	200	108	49	6	135	99	141	29	9

¹Zimmerman R D, Murillo-Sánchez C E, Thomas R J. MATPOWER: Steady-state operations, planning, and analysis tools for power systems research and education[J]. IEEE Transactions on Power Systems, 2010, 26(1): 12–19.

²Schmidt M, Aßmann D, Burlacu R, et al. GasLib—A library of gas network instances[J]. Data, 2017, 2(4): 40.

Table 3: Reinforcement-Learning Policy-State, DCV-State, and Action Dimensions.

Case	Policy state dimension	DCV state dimension	Action-space dimension
P14_G11	33	48	15
P57_G40	152	173	23
P200_G135	456	564	119

The policy-state dimensions in Table 3 refer to the observation vectors passed to FOCOPS, consisting of nonzero electric-load active/reactive-power profiles, gas-load profiles, and dynamic line-pack terms. The DCV-state dimensions refer to the separate constraint-monitoring vector used in the safety cost: external-grid active/reactive power, online-generator reactive-power outputs, bus-voltage magnitudes, line currents, slack-well production, gas-node pressures, and electric-compressor power consumptions. The normalized action vector is mapped to physical controls including online-generator active-power set points, online-generator-bus and reference-bus voltage set points, P2G electric-power consumptions, non-slack gas-well productions, gas-reference-node pressure set points, and compressor ratios. For P200_G135, generators with `net.gen.in_service=False` are excluded from the action vector, the generator-related DCV entries, and the polynomial generator-cost calculation, consistent with the standalone power-system environments; the policy-state dimension is unchanged because it does not contain generator-indexed terms.

4 Engineering-Unit Coefficients and Electric–Gas Coupling Conversions

4.1 Pipeline Model and Variables

For ordinary pipelines, the dynamic line-pack term and the steady-state flow term are characterized by coefficients K_m and K_q , respectively. For ordinary pipeline ℓ , let X_ℓ denote the pipeline length (m), D_ℓ the pipe diameter (m), $Z = 0.8$ the compressibility factor, $T = 281.15$ K the operating temperature, $G_s = 0.6106$ the relative density of natural gas, and $\epsilon = 0.05$ the pipe roughness parameter. Gas pressure is expressed in bar and gas flow is expressed in MSm^3/h .

To clarify the origin of these two coefficients, consider the simplified transient model of a horizontal gas transmission pipeline. After neglecting elevation and inertial terms, the continuity and momentum equations can be written as

$$\frac{\partial(\rho v)}{\partial x} + \frac{\partial \rho}{\partial t} = 0, \quad (1a)$$

$$\frac{\partial p}{\partial x} + \frac{f_\ell^{\text{fr}} \rho v |v|}{2D_\ell} = 0, \quad (1b)$$

where ρ is the gas density, v is the in-pipe gas velocity, p is pressure, and f_ℓ^{fr} is the friction factor. Combining the gas equation of state with the standard-state volumetric-flow definition gives

$$p = \rho R T Z, \quad (2a)$$

$$\tilde{f}_\ell = \frac{\pi D_\ell^2}{4\rho_0} \rho v, \quad (2b)$$

where R is the gas constant, ρ_0 is the standard-state gas density, and $\tilde{f}_\ell$ is the standard-state volumetric flow.

The pressure-flow form of the pipeline equations is then obtained as

$$\frac{\partial \tilde{f}_\ell}{\partial x} + \frac{\pi D_\ell^2}{4\rho_0 RT Z} \frac{\partial p}{\partial t} = 0, \quad (3a)$$

$$p \frac{\partial p}{\partial x} + \left(\frac{4}{\pi}\right)^2 \frac{f_\ell^{\text{fr}} RT Z \rho_0^2}{2D_\ell^5} \tilde{f}_\ell |\tilde{f}_\ell| = 0. \quad (3b)$$

The factor $\pi/4$ originates from the circular cross-sectional area $A_\ell = \pi D_\ell^2/4$. In the flow equation, because velocity is expressed as $v \propto \tilde{f}_\ell/A_\ell$ and the friction term contains $v|v|$, the squared-area coefficient associated with $(\pi/4)^2$ appears.

4.2 Line-Pack and Weymouth-Flow Terms

Discretizing pipeline $\ell = (i, j)$ over its length X_ℓ , and representing the pipe state by the endpoint-pressure average $\bar{p}_{\ell,t} = (p_{i,t} + p_{j,t})/2$ and the average flow $\tilde{f}_{\ell,t} = (F_{\ell,t}^{\text{in}} + F_{\ell,t}^{\text{out}})/2$, yields the line-pack term and the Weymouth-type flow term:

$$m_{\ell,t} = \Phi_\ell^m \frac{\pi X_\ell D_\ell^2}{4RT Z \rho_0} \bar{p}_{\ell,t}, \quad (4a)$$

$$m_{\ell,t} = m_{\ell,t-1} + F_{\ell,t}^{\text{in}} - F_{\ell,t}^{\text{out}}, \quad (4b)$$

$$\tilde{f}_{\ell,t} |\tilde{f}_{\ell,t}| = \Phi_\ell^q \left(\frac{\pi}{4}\right)^2 \frac{D_\ell^5}{X_\ell f_\ell^{\text{fr}} RT Z \rho_0^2} (p_{i,t}^2 - p_{j,t}^2). \quad (4c)$$

Here, Φ_ℓ^m and Φ_ℓ^q collect the conversions among pressure, time, and standard-volume units. If $\Sigma_{\ell,t} = F_{\ell,t}^{\text{in}} + F_{\ell,t}^{\text{out}}$ is introduced, then $\tilde{f}_{\ell,t} = \Sigma_{\ell,t}/2$. Therefore, the Weymouth relation in average-flow form can equivalently be written as $\Sigma_{\ell,t} |\Sigma_{\ell,t}| = 4K_{q,\ell} (p_{i,t}^2 - p_{j,t}^2)$.

4.3 Engineering-Unit Coefficients

Under the engineering units adopted in this document, the above line-pack and flow terms are folded into two tabulated coefficients, $K_{m,\ell}$ and $K_{q,\ell}$. The friction correction term is first computed as³

$$\Lambda_\ell = \left[2 \log_{10} \left(\frac{3.7 D_\ell}{\epsilon} \right) \right]^2, \quad (5)$$

where Λ_ℓ can be interpreted as an approximate expression for the reciprocal of the Darcy friction factor. After combining $1/f_\ell^{\text{fr}}$, standard-state parameters, and unit conversions, the pipeline coefficients are written as

$$K_{m,\ell} = C_m \frac{X_\ell D_\ell^2}{TZ}, \quad K_{q,\ell} = C_q \frac{D_\ell^5 \Lambda_\ell}{ZT X_\ell G_s}. \quad (6)$$

The folded constants used in this document are

$$C_m = 2.119 \times 10^{-7}, \quad C_q = 96.074830 \times 10^{-15}. \quad (7)$$

These two numerical constants result from combining standard-state parameters, pressure units, time units, and volume scaling. Taking the standard temperature as $T_0 = 273.15$ K, the standard pressure as $p_0 = 1.01325$ bar, and the air density as $\rho_{\text{air}} = 1.2922$ kg/m³, the line-pack constant can be obtained from the ideal-gas equation of state and the circular pipe cross section:

$$C_m \simeq \frac{\pi T_0}{4 p_0} \times 10^{-9} = \frac{\pi}{4} \frac{273.15}{1.01325} \times 10^{-9} \approx 2.12 \times 10^{-7}, \quad (8)$$

where 10^{-9} accounts for the consistent conversion from standard volume to million standard cubic meters and for the length and area scales.

³Humpola J. Gas network optimization by MINLP[M]. Logos Verlag Berlin GmbH, 2017; De Wolf D, Smeers Y. The gas transmission problem solved by an extension of the simplex algorithm[J]. Management Science, 2000, 46(11): 1454–1465.

The flow-term constant C_q follows from the Weymouth-type pipeline-flow relation. When the squared-flow term is expressed in MSm^3/h and the squared-pressure term in bar^2 , the part independent of pipe diameter, length, and friction can be written as

$$C_q = \left(\frac{\pi}{4}\right)^2 10^5 (3600)^2 \frac{T_0}{p_0 \rho_{\text{air}}} s_q, \quad (9)$$

where $(\pi/4)^2$ originates from the square of the circular cross-sectional area, 10^5 converts the pressure scale between bar and Pa, $(3600)^2$ converts squared flow from seconds to hours, $T_0/(p_0 \rho_{\text{air}})$ converts between standard-state volume and mass density, and s_q collects the MSm^3 volume scale, the squared-pressure scale, and the length-diameter unit convention adopted by the gas-network data. Under the unit system used here,

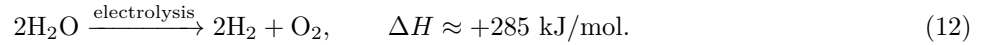
$$s_q = 5.7606303 \times 10^{-28}, \quad (10)$$

therefore

$$C_q = \left(\frac{\pi}{4}\right)^2 10^5 (3600)^2 \frac{273.15}{1.01325 \times 1.2922} \times 5.7606303 \times 10^{-28} \approx 96.074830 \times 10^{-15}. \quad (11)$$

Thus, the tabulated K_m and K_q values are the pipeline coefficients obtained from the line-pack and Weymouth flow terms under the above unit system, given X_ℓ , D_ℓ , and Λ_ℓ .

For the power-to-gas (P2G) coupling element, the electric-to-gas conversion is represented by a lumped efficiency that can be interpreted from the standard two-stage P2G route.⁴ In the first stage, water electrolysis converts electric energy into hydrogen:



In the second stage, the Sabatier methanation reaction converts hydrogen and carbon dioxide into synthetic methane:



Let η_{el} denote the electrolysis-stage efficiency and η_{met} denote the effective methanation-stage efficiency, including the auxiliary losses associated with conversion and conditioning. The lumped P2G efficiency used in the coupling tables is then

$$\eta_{\text{p2g}} = \eta_{\text{el}} \eta_{\text{met}}. \quad (14)$$

Based on the above two-stage conversion route, the overall P2G efficiency is commonly taken in the range from 0.4 to 0.6. The value reported in the P2G tables is therefore a representative benchmark setting within this range. Accordingly, the gas output of a P2G unit is modeled as proportional to its electric input:

$$F_t^{\text{p2g}} = \Phi_g \eta_{\text{p2g}} P_t^{\text{p2g}}, \quad (15)$$

where P_t^{p2g} is the consumed electric power, F_t^{p2g} is the injected synthetic-natural-gas flow, and Φ_g is the engineering-unit conversion coefficient between electric energy and standard gas volume.

4.4 Electric-Gas Energy Conversion

It should be distinguished that the energy-volume conversion coefficients for electric-gas coupling elements are based on a separate set of EIA energy-unit data⁵, which is used to relate per-unit electric power to natural-gas volume. The simplified conversion adopted here is

$$1 \text{ MWh} \approx 3412.1416 \text{ ft}^3 \times 0.02831685 \approx 96.6 \text{ m}^3, \quad (16)$$

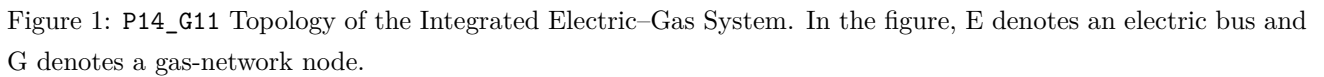
⁴Comparable P2G conversion-efficiency formulations are used in Clegg S, Mancarella P. Integrated modeling and assessment of the operational impact of power-to-gas (P2G) on electrical and gas transmission networks[J]. IEEE Transactions on Sustainable Energy, 2015, 6(4): 1234–1244; and He C, Wu L, Liu T, et al. Robust co-optimization planning of interdependent electricity and natural gas systems with a joint N-1 and probabilistic reliability criterion[J]. IEEE Transactions on Power Systems, 2017, 33(2): 2140–2154.

⁵EIA – Units & Calculators Explained, <https://www.eia.gov/energyexplained/units-and-calculators/>.

$$\alpha_0 = \frac{96.6 \times 100}{10^6} = 9.66 \times 10^{-3} \text{ MSm}^3/(100 \text{ MWh}). \quad (17)$$

5 P14_G11

P14_G11 consists of the IEEE 14-bus power network and an 11-node gas network. The power network is constructed using `pandapower.networks.case14()`; the 11-node gas network is a purpose-built small synthetic pipeline network for testing, containing three gas wells, three load nodes, eight ordinary pipelines, two compressors, and one P2G unit.



5.2 Gas-Side Data

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Gas["Well"]: Gas-Well Data

Table 4: P14_G11 Gas-Well Data.

No.	$G_{\max}$ MSm ³ /h	$G_{\min}$ MSm ³ /h	Cost coeff.	Ref. well	Output MSm ³ /h
1	0.16	0	228.566	1	0
2	0.115	0	190.472	0	0
3	1	0	355.548	0	0

Gas["Load"]: Gas-Network Node Data

Table 5: P14_G11 Gas-Network Node Data.

No.	$p_{\max}$ bar	$p_{\min}$ bar	Load MSm ³ /h	Type	p_{ref} ⁶ bar
1	70	40	0	1	50
2	70	40	0	3	0
3	70	40	0	3	0
4	70	40	0.1	2	0
5	60	40	0.12	2	0
6	60	40	0.08	2	0
7	70	40	0	0	0
8	70	40	0	0	0
9	70	40	0	0	0
10	70	40	0	0	0
11	70	40	0	0	0

Gas["Pipeline"]: Gas-network Pipeline Data

Table 6: P14_G11 Gas-Network Pipeline Data.

No.	in	out	Len. m	D m	K_m MSm ³ /(h Pa)	K_q MSm ⁶ /(h ² bar ²)
1	1	2	55000	0.5	1.2943e−2	6.8999e−5
2	7	8	55000	0.5	1.2943e−2	6.8999e−5
3	3	9	55000	0.5	1.2943e−2	6.8999e−5
4	8	4	55000	0.5	1.2943e−2	6.8999e−5
5	8	10	55000	0.5	1.2943e−2	6.8999e−5
6	9	10	55000	0.5	1.2943e−2	6.8999e−5
7	11	5	55000	0.5	1.2943e−2	6.8999e−5
8	11	6	55000	0.5	1.2943e−2	6.8999e−5

⁶ p_{ref} denotes the prescribed pressure of the gas reference node. For non-reference nodes, this field is listed as zero because their operating pressures are determined by the gas-flow equations.

5.3 Power-Side Units

This subsection lists all power-side generation units, including the reference-bus generator and the GT units coupled to the gas network.

Table 7: P14_G11 Power-Side Generation Units.

No.	Power bus	Type	Gas node	Eff. p.u.	$V_{g,\min}$ p.u.	$V_{g,\max}$ p.u.	$P_{\min}$ MW	$P_{\max}$ MW	$Q_{\min}$ MVA _r	$Q_{\max}$ MVA _r
1	1	GP	—	—	0.94	1.06	0.00	332.40	0.00	12.00
2	2	GP	—	—	0.94	1.06	14.00	140.00	-40.00	50.00
3	3	GT	4	0.55	0.94	1.06	14.00	100.00	0.00	40.00
4	6	GT	5	0.45	0.94	1.06	0.00	100.00	-6.00	24.00
5	8	GP	—	—	0.94	1.06	0.00	100.00	-6.00	24.00

5.4 Electric-Gas Coupling Elements

This subsection reports P2G units and compressors that couple the electric and gas subsystems.

Table 8: P14_G11 P2G Units.

No.	Power bus	Gas node	Eff. ⁷ p.u.	$P_{\min}$ MW	$P_{\max}$ MW	Coupling strength ⁸
1	4	6	0.45	0.00	8.00	750.00

Table 9: P14_G11 Compressor Units; E and G denote electric-driven and gas-driven compressors, respectively.

No.	In.	Out.	Drv.	Power bus	Ratio p.u.	Power MW
1	2	7	E	6	1.00–1.20	0–20
2	10	11	G	6	1.00–1.20	—

6 P57_G40

6.1 Construction Summary

P57_G40 consists of the IEEE 57-bus power network and a 40-node gas network. This 40-node gas network roughly represents a segment of Germany’s low-calorific gas transmission network serving the Rhine-Main-Ruhr region. The network contains three gas wells, 29 load nodes, 39 ordinary pipelines, six compressors, and one P2G unit.

The topology in Fig. 2 is used as the reference map for interpreting the following data tables. The gas-side tables enumerate the source nodes, demand nodes, and pipeline parameters, while the power-side and coupling tables summarize the generators, gas turbines, P2G unit, and compressors associated with this integrated case.

⁷The P2G efficiency column reports the lumped electric-to-gas efficiency η_{p2g} defined in Section 4.3.

⁸The coupling-strength coefficient is a benchmark scaling factor used to tune the gas-side impact of P2G injection under the adopted engineering-unit conversion. It is not an additional conversion efficiency.

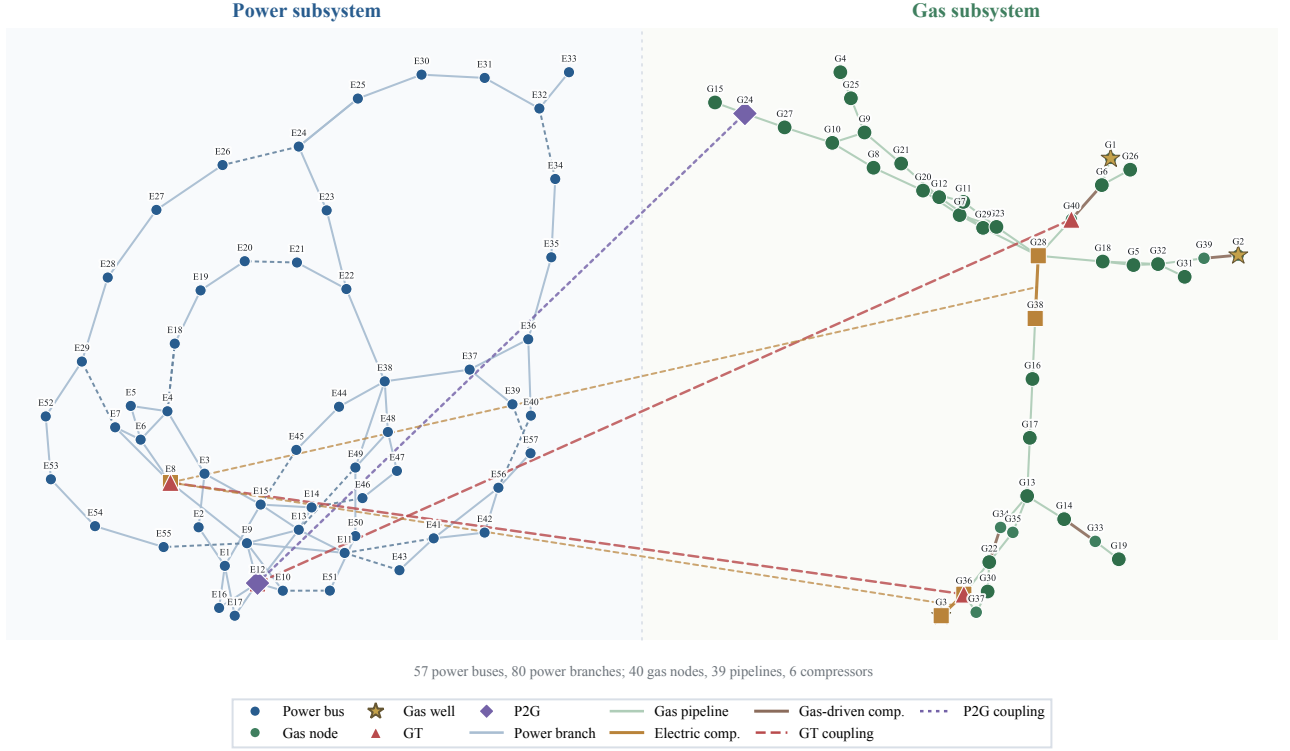


Figure 2: P57_G40 Topology of the Integrated Electric-Gas System. In the figure, E denotes an electric bus and G denotes a gas-network node.

6.2 Gas-Side Data

This subsection reports the gas-side benchmark tables `Gas["Well"]`, `Gas["Load"]`, and `Gas["Pipeline"]`. The table `Gas["Load"]` corresponds to the gas-network node data. In the node-type flag, 1 denotes a reference node, 2 a gas-load node, 3 a gas-source node, and 0 an ordinary junction node.

`Gas["Well"]`: Gas-Well Data

Table 10: P57_G40 Gas-Well Data.

No.	$G_{\max}$ MSm ³ /h	$G_{\min}$ MSm ³ /h	Cost coeff.	Ref. well	Output MSm ³ /h
1	2.0	0	2692.34	1	0
2	0.1	0	2243.61	0	0
3	0.5	0	2243.61	0	0

`Gas["Load"]`: Gas-Network Node Data

Table 11: P57_G40 Gas-Network Node Data.

No.	$p_{\max}$ bar	$p_{\min}$ bar	Load MSm ³ /h	Type	p_{ref} bar	No.	$p_{\max}$ bar	$p_{\min}$ bar	Load MSm ³ /h	Type	p_{ref} bar
1	81.0132	68	0	1	74	21	81.0132	30	0.075	2	0
2	81.0132	30	0	3	0	22	81.0132	30	0.075	2	0
3	81.0132	30	0	3	0	23	81.0132	30	0.075	2	0

No.	$p_{\max}$ bar	$p_{\min}$ bar	Load MSm ³ /h	Type	p_{ref} bar	No.	$p_{\max}$ bar	$p_{\min}$ bar	Load MSm ³ /h	Type	p_{ref} bar
4	81.0132	30	0.075	2	0	24	81.0132	30	0.075	2	0
5	81.0132	30	0.075	2	0	25	81.0132	30	0.075	2	0
6	81.0132	30	0.075	2	0	26	81.0132	30	0.075	2	0
7	81.0132	30	0.075	2	0	27	81.0132	30	0.075	2	0
8	81.0132	30	0.075	2	0	28	81.0132	30	0.075	2	0
9	81.0132	30	0.075	2	0	29	81.0132	30	0.075	2	0
10	81.0132	30	0.075	2	0	30	81.0132	30	0.075	2	0
11	81.0132	30	0.075	2	0	31	81.0132	30	0.075	2	0
12	81.0132	30	0.075	2	0	32	81.0132	30	0.075	2	0
13	81.0132	30	0.075	2	0	33	81.0132	30	0	0	0
14	81.0132	30	0.075	2	0	34	81.0132	30	0	0	0
15	81.0132	30	0.075	2	0	35	81.0132	30	0	0	0
16	81.0132	30	0.075	2	0	36	81.0132	30	0	0	0
17	81.0132	30	0.075	2	0	37	81.0132	30	0	0	0
18	81.0132	30	0.075	2	0	38	81.0132	30	0	0	0
19	81.0132	30	0.075	2	0	39	81.0132	30	0	0	0
20	81.0132	30	0.075	2	0	40	81.0132	30	0	0	0

Gas["Pipeline"]: Gas-network Pipeline Data

Table 12: P57_G40 Gas-Network Pipeline Data.

No.	in	out	Len. m	D m	K_m MSm ³ /(h Pa)	K_q MSm ⁶ /(h ² bar ²)	No.	in	out	Len. m	D m	K_m MSm ³ /(h Pa)	K_q MSm ⁶ /(h ² bar ²)
1	1	6	13071	1	1.2304e−2	9.2906e−3	21	20	7	10586	0.6	3.5874e−3	8.9203e−4
2	33	19	38447	0.8	2.3163e−2	1.0350e−3	22	20	11	10452	0.6	3.5420e−3	9.0347e−4
3	38	16	21558	1	2.0293e−2	5.6331e−3	23	6	26	12397	0.8	7.4687e−3	3.2099e−3
4	16	17	6998	1	6.5875e−3	1.7353e−2	24	11	23	19303	0.6	6.5414e−3	4.8920e−4
5	17	13	58219	0.8	3.5074e−2	6.8350e−4	25	28	23	66037	0.6	2.2379e−2	1.4300e−4
6	28	29	43345	0.8	2.6114e−2	9.1805e−4	26	28	18	18969	1	1.7856e−2	6.4019e−3
7	29	12	16579	0.6	5.6183e−3	5.6958e−4	27	18	32	36061	0.8	2.1725e−2	1.1035e−3
8	12	21	10023	0.6	3.3966e−3	9.4214e−4	28	32	31	22224	0.8	1.3389e−2	1.7905e−3
9	29	7	35219	0.6	1.1935e−2	2.6812e−4	29	32	5	31180	0.8	1.8785e−2	1.2762e−3
10	7	23	20322	0.6	6.8868e−3	4.6467e−4	30	5	18	12767	1	1.2018e−2	9.5119e−3
11	21	9	32868	0.8	1.9802e−2	1.2107e−3	31	32	39	32921	0.8	1.9834e−2	1.2087e−3
12	28	40	47488	0.8	2.8609e−2	8.3796e−4	32	36	22	49866	0.8	3.0042e−2	7.9799e−4
13	9	10	3803	0.6	1.2888e−3	2.4830e−3	33	22	35	3479	0.8	2.0959e−3	1.1438e−2
14	9	25	39036	1.8	1.1906e−1	5.8783e−2	34	36	37	3418	1	3.2175e−3	3.5529e−2
15	10	27	38660	1.8	1.1791e−1	5.9355e−2	35	30	37	32449	1	3.0546e−2	3.7424e−3
16	25	4	18018	1.8	5.4954e−2	1.2735e−1	36	30	22	26427	0.8	1.5921e−2	1.5058e−3
17	27	24	3068	1.8	9.3572e−3	7.4793e−1	37	13	14	18137	1	1.7073e−2	6.6956e−3

No.	in	out	Len. m	D m	K_m MSm ³ /(h Pa)	K_q MSm ⁶ /(h ² bar ²)	No.	in	out	Len. m	D m	K_m MSm ³ /(h Pa)	K_q MSm ⁶ /(h ² bar ²)
18	24	15	12016	1.8	3.6648e−2	1.9097e−1	38	13	34	32528.5	0.8	1.9597e−2	1.2233e−3
19	10	8	14043	0.4	2.1151e−3	8.8551e−5	39	13	35	32766	0.8	1.9740e−2	1.2145e−3
20	8	20	20635	0.6	6.9928e−3	4.5762e−4							

6.3 Power-Side Units

This subsection lists all power-side generation units, including the reference-bus generator and the GT units coupled to the gas network.

Table 13: P57_G40 Power-Side Generation Units.

No.	Power bus	Type	Gas node	Eff. p.u.	$V_{g,\min}$ p.u.	$V_{g,\max}$ p.u.	$P_{\min}$ MW	$P_{\max}$ MW	$Q_{\min}$ MVA _r	$Q_{\max}$ MVA _r
1	1	GP	—	—	0.94	1.06	0.00	575.88	-140.00	200.00
2	2	GP	—	—	0.94	1.06	0.00	100.00	-17.00	50.00
3	3	GP	—	—	0.94	1.06	0.00	140.00	-10.00	60.00
4	6	GP	—	—	0.94	1.06	0.00	100.00	-8.00	25.00
5	8	GT	36	0.55	0.94	1.06	0.00	550.00	-140.00	200.00
6	9	GP	—	—	0.94	1.06	0.00	100.00	-3.00	9.00
7	12	GT	40	0.45	0.94	1.06	0.00	410.00	-150.00	155.00

6.4 Electric-Gas Coupling Elements

This subsection reports P2G units and compressors that couple the electric and gas subsystems.

Table 14: P57_G40 P2G Units.

No.	Power bus	Gas node	Eff. p.u.	$P_{\min}$ MW	$P_{\max}$ MW	Coupling strength
1	12	24	0.45	70.00	150.00	50.00

Table 15: P57_G40 Compressor Units; E and G denote electric-driven and gas-driven compressors, respectively.

No.	In.	Out.	Drv.	Power bus	Ratio p.u.	Power MW
1	38	28	E	8	1.00–1.20	0–140
2	14	33	G	8	1.00–1.20	—
3	22	34	G	12	1.00–1.20	—
4	3	36	E	8	1.00–1.20	0–10
5	2	39	G	8	1.00–1.20	—
6	6	40	G	12	1.00–1.20	—

7 P200_G135

7.1 Construction Summary

P200_G135 consists of the ACTIVSg200 power network and a 135-node gas network. This 135-node gas network roughly represents a portion of Germany's high-calorific gas transmission network. The network contains six GT units, nine P2G units, and 29 compressors.

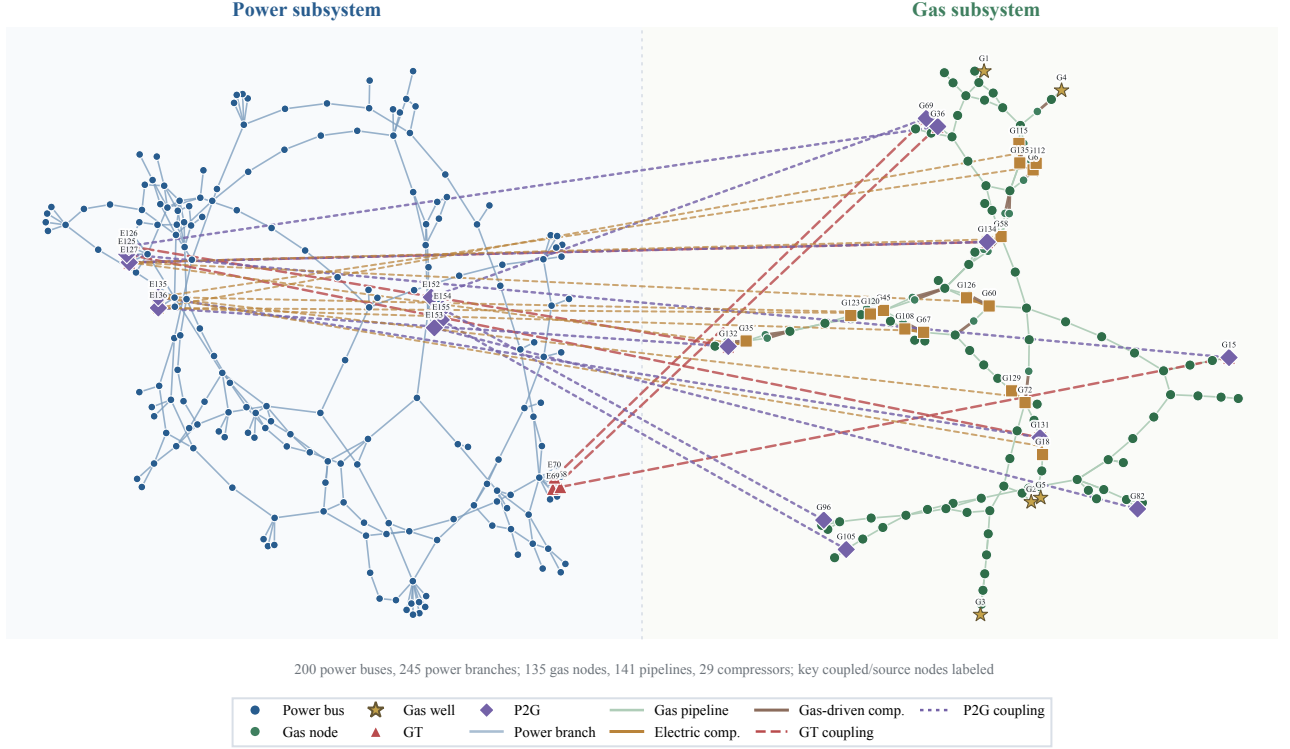


Figure 3: P200_G135 Topology of the Integrated Electric-Gas System. In the figure, E denotes an electric bus and G denotes a gas-network node; only coupled nodes and gas-source nodes are labeled to avoid visual clutter.

The topology in Fig. 3 gives the high-level layout of the largest benchmark. Because the graph is dense, the following tables serve as the detailed data specification: gas-side tables list the wells, nodes, and pipelines, and the power-side and coupling tables report the generation units, P2G units, and compressors that define the electric-gas interfaces.

7.2 Gas-Side Data

This subsection reports the gas-side benchmark tables `Gas["Well"]`, `Gas["Load"]`, and `Gas["Pipeline"]`. The table `Gas["Load"]` corresponds to the gas-network node data. In the node-type flag, 1 denotes a reference node, 2 a gas-load node, 3 a gas-source node, and 0 an ordinary junction node.

`Gas["Well"]`: Gas-Well Data

Table 16: P200_G135 Gas-Well Data.

No.	$G_{\max}$ MSm ³ /h	$G_{\min}$ MSm ³ /h	Cost coeff.	Ref. well	Output MSm ³ /h
1	2	0	828	1	0
2	2	0	690	0	0

No.	$G_{\max}$ MSm ³ /h	$G_{\min}$ MSm ³ /h	Cost coeff.	Ref. well	Output MSm ³ /h
3	2	0	690	0	0
4	2	0	690	0	0
5	2	0	690	0	0
6	2	0	690	0	0

Gas["Load"]: Gas-Network Node Data

Table 17: P200_G135 Gas-Network Node Data.

No.	$p_{\max}$ bar	$p_{\min}$ bar	Load MSm ³ /h	Type	p_{ref} bar	No.	$p_{\max}$ bar	$p_{\min}$ bar	Load MSm ³ /h	Type	p_{ref} bar
1	75	60	0	1	60	69	81.0132	30	0.04	2	0
2	81.0132	30	0	3	0	70	81.0132	30	0.04	2	0
3	81.0132	30	0	3	0	71	81.0132	30	0.04	2	0
4	81.0132	30	0	3	0	72	81.0132	30	0.04	2	0
5	81.0132	30	0	3	0	73	81.0132	30	0.04	2	0
6	81.0132	30	0	3	0	74	81.0132	30	0.04	2	0
7	81.0132	30	0.04	2	0	75	81.0132	30	0.04	2	0
8	81.0132	30	0.04	2	0	76	81.0132	30	0.04	2	0
9	81.0132	30	0.04	2	0	77	81.0132	30	0.04	2	0
10	81.0132	30	0.04	2	0	78	81.0132	30	0.04	2	0
11	81.0132	30	0.04	2	0	79	81.0132	30	0.04	2	0
12	81.0132	30	0.04	2	0	80	81.0132	30	0.04	2	0
13	81.0132	30	0.04	2	0	81	81.0132	30	0.04	2	0
14	81.0132	30	0.04	2	0	82	81.0132	30	0.04	2	0
15	81.0132	30	0.04	2	0	83	81.0132	30	0.04	2	0
16	81.0132	30	0.04	2	0	84	81.0132	30	0.04	2	0
17	81.0132	30	0.04	2	0	85	81.0132	30	0.04	2	0
18	81.0132	30	0.04	2	0	86	81.0132	30	0.04	2	0
19	81.0132	30	0.04	2	0	87	81.0132	30	0.04	2	0
20	81.0132	30	0.04	2	0	88	81.0132	30	0.04	2	0
21	81.0132	30	0.04	2	0	89	81.0132	30	0.04	2	0
22	81.0132	30	0.04	2	0	90	81.0132	30	0.04	2	0
23	81.0132	30	0.04	2	0	91	81.0132	30	0.04	2	0
24	81.0132	30	0.04	2	0	92	81.0132	30	0.04	2	0
25	81.0132	30	0.04	2	0	93	81.0132	30	0.04	2	0
26	81.0132	30	0.04	2	0	94	81.0132	30	0.04	2	0
27	81.0132	30	0.04	2	0	95	81.0132	30	0.04	2	0
28	81.0132	30	0.04	2	0	96	81.0132	30	0.04	2	0
29	81.0132	30	0.04	2	0	97	81.0132	30	0.04	2	0
30	81.0132	30	0.04	2	0	98	81.0132	30	0.04	2	0

No.	$p_{\max}$ bar	$p_{\min}$ bar	Load MSm ³ /h	Type	p_{ref} bar	No.	$p_{\max}$ bar	$p_{\min}$ bar	Load MSm ³ /h	Type	p_{ref} bar
31	81.0132	30	0.04	2	0	99	81.0132	30	0.04	2	0
32	81.0132	30	0.04	2	0	100	81.0132	30	0.04	2	0
33	81.0132	30	0.04	2	0	101	81.0132	30	0.04	2	0
34	81.0132	30	0.04	2	0	102	81.0132	30	0.04	2	0
35	81.0132	30	0.04	2	0	103	81.0132	30	0.04	2	0
36	81.0132	30	0.04	2	0	104	81.0132	30	0.04	2	0
37	81.0132	30	0.04	2	0	105	81.0132	30	0.04	2	0
38	81.0132	30	0.04	2	0	106	81.0132	30	0	0	0
39	81.0132	30	0.04	2	0	107	81.0132	30	0	0	0
40	81.0132	30	0.04	2	0	108	81.0132	30	0	0	0
41	81.0132	30	0.04	2	0	109	81.0132	30	0	0	0
42	81.0132	30	0.04	2	0	110	81.0132	30	0	0	0
43	81.0132	30	0.04	2	0	111	81.0132	30	0	0	0
44	81.0132	30	0.04	2	0	112	81.0132	30	0	0	0
45	81.0132	30	0.04	2	0	113	81.0132	30	0	0	0
46	81.0132	30	0.04	2	0	114	81.0132	30	0	0	0
47	81.0132	30	0.04	2	0	115	81.0132	30	0	0	0
48	81.0132	30	0.04	2	0	116	81.0132	30	0	0	0
49	81.0132	30	0.04	2	0	117	81.0132	30	0	0	0
50	81.0132	30	0.04	2	0	118	81.0132	30	0	0	0
51	81.0132	30	0.04	2	0	119	81.0132	30	0	0	0
52	81.0132	30	0.04	2	0	120	81.0132	30	0	0	0
53	81.0132	30	0.04	2	0	121	81.0132	30	0	0	0
54	81.0132	30	0.04	2	0	122	81.0132	30	0	0	0
55	81.0132	30	0.04	2	0	123	81.0132	30	0	0	0
56	81.0132	30	0.04	2	0	124	81.0132	30	0	0	0
57	81.0132	30	0.04	2	0	125	81.0132	30	0	0	0
58	81.0132	30	0.04	2	0	126	81.0132	30	0	0	0
59	81.0132	30	0.04	2	0	127	81.0132	30	0	0	0
60	81.0132	30	0.04	2	0	128	81.0132	30	0	0	0
61	81.0132	30	0.04	2	0	129	81.0132	30	0	0	0
62	81.0132	30	0.04	2	0	130	81.0132	30	0	0	0
63	81.0132	30	0.04	2	0	131	81.0132	30	0	0	0
64	81.0132	30	0.04	2	0	132	81.0132	30	0	0	0
65	81.0132	30	0.04	2	0	133	81.0132	30	0	0	0
66	81.0132	30	0.04	2	0	134	81.0132	30	0	0	0
67	81.0132	30	0.04	2	0	135	81.0132	30	0	0	0
68	81.0132	30	0.04	2	0						

Gas["Pipeline"]: Gas-network Pipeline Data

Table 18: P200_G135 Gas-Network Pipeline Data.

No.	in	out	Len. m	D m	K_m MSm ³ /(h Pa)	K_q MSm ⁶ /(h ² bar ²)	No.	in	out	Len. m	D m	K_m MSm ³ /(h Pa)	K_q MSm ⁶ /(h ² bar ²)
1	3	28	50304.8	1.2	6.8190e−2	6.0069e−3	72	39	60	56432.5	1.2	7.6496e−2	5.3547e−3
2	131	25	50170.3	1.2	6.8007e−2	6.0230e−3	73	108	63	79066.7	1.2	1.0718e−1	3.8218e−3
3	90	91	17722	1.2	2.4023e−2	1.7051e−2	74	107	45	84957.9	1.2	1.1516e−1	3.5568e−3
4	90	91	17722	1.2	2.4023e−2	1.7051e−2	75	63	45	99697	1.2	1.3514e−1	3.0310e−3
5	91	38	27576	1.2	3.7380e−2	1.0958e−2	76	2	26	42281.5	1.2	5.7314e−2	7.1468e−3
6	90	7	50039.1	1.2	6.7829e−2	6.0388e−3	77	45	56	92797.1	1.2	1.2579e−1	3.2563e−3
7	5	2	35005.3	1.2	4.7451e−2	8.6323e−3	78	120	42	24981.3	1.2	3.3863e−2	1.2096e−2
8	28	51	35640.1	1.2	4.8311e−2	8.4786e−3	79	121	42	25743.4	1.2	3.4896e−2	1.1738e−2
9	51	38	12536.2	1.2	1.6993e−2	2.4104e−2	80	45	124	84458.2	1.2	1.1449e−1	3.5778e−3
10	101	105	60939.7	1.2	8.2606e−2	4.9586e−3	81	45	125	84946.7	1.2	1.1515e−1	3.5573e−3
11	105	98	64597.2	1.2	8.7563e−2	4.6779e−3	82	126	31	80621.8	1.2	1.0929e−1	3.7481e−3
12	98	99	23354.6	1.2	3.1658e−2	1.2939e−2	83	127	31	80904.3	1.2	1.0967e−1	3.7350e−3
13	71	7	61293	1.2	8.3084e−2	4.9300e−3	84	122	46	83083	1.2	1.1262e−1	3.6370e−3
14	99	30	43259.3	1.2	5.8639e−2	6.9852e−3	85	123	46	83957.7	1.2	1.1381e−1	3.5992e−3
15	30	86	39290.8	1.2	5.3260e−2	7.6908e−3	86	46	64	13264.2	1.2	1.7980e−2	2.2781e−2
16	86	87	38503.4	1.2	5.2192e−2	7.8481e−3	87	26	78	49223.1	1.2	6.6723e−2	6.1389e−3
17	87	80	65028.4	1.2	8.8148e−2	4.6468e−3	88	46	64	13264.2	1.2	1.7980e−2	2.2781e−2
18	96	17	81181.6	1.2	1.1004e−1	3.7222e−3	89	110	35	87864.1	1.2	1.1910e−1	3.4391e−3
19	82	81	6510.82	1.2	8.8256e−3	4.6411e−2	90	109	35	87454.8	1.2	1.1855e−1	3.4552e−3
20	81	89	23219.5	1.2	3.1475e−2	1.3014e−2	91	132	70	33438.2	1.2	4.5326e−2	9.0369e−3
21	89	94	69525.7	1.2	9.4244e−2	4.3463e−3	92	133	70	33473.9	1.2	4.5375e−2	9.0272e−3
22	89	94	69525.7	1.2	9.4244e−2	4.3463e−3	93	39	53	42519.4	1.2	5.7636e−2	7.1068e−3
23	94	93	29067.2	1.2	3.9401e−2	1.0396e−2	94	53	68	21281.5	1.2	2.8848e−2	1.4199e−2
24	71	73	35220.9	1.2	4.7743e−2	8.5795e−3	95	39	59	37246.5	1.2	5.0489e−2	8.1129e−3
25	93	80	73550.2	1.2	9.9699e−2	4.1084e−3	96	59	58	63409.2	1.2	8.5953e−2	4.7655e−3
26	94	104	46081.6	1.2	6.2465e−2	6.5574e−3	97	31	32	13796.9	1.2	1.8702e−2	2.1902e−2
27	104	88	49928.8	1.2	6.7680e−2	6.0522e−3	98	5	78	10397.6	1.2	1.4094e−2	2.9062e−2
28	94	102	85858	1.2	1.1638e−1	3.5195e−3	99	31	32	13796.9	1.2	1.8702e−2	2.1902e−2
29	94	102	85858	1.2	1.1638e−1	3.5195e−3	100	31	32	13796.9	1.2	1.8702e−2	2.1902e−2
30	102	11	10634.5	1.2	1.4415e−2	2.8415e−2	101	31	40	14097.1	1.2	1.9109e−2	2.1435e−2
31	102	11	10634.5	1.2	1.4415e−2	2.8415e−2	102	31	40	14097.1	1.2	1.9109e−2	2.1435e−2
32	80	66	68792.3	1.2	9.3250e−2	4.3926e−3	103	32	116	81461.5	1.2	1.1042e−1	3.7094e−3
33	102	85	37563.1	1.2	5.0918e−2	8.0445e−3	104	32	117	82408.8	1.2	1.1171e−1	3.6668e−3
34	85	83	74204.5	1.2	1.0059e−1	4.0722e−3	105	32	34	47648.2	1.2	6.4589e−2	6.3418e−3
35	73	21	44307.6	1.2	6.0060e−2	6.8200e−3	106	34	134	37524.1	1.2	5.0865e−2	8.0529e−3
36	83	79	34864.5	1.2	4.7260e−2	8.6672e−3	107	58	55	55637.1	1.2	7.5418e−2	5.4312e−3
37	97	79	10819	1.2	1.4666e−2	2.7930e−2	108	58	118	89446.2	1.2	1.2125e−1	3.3783e−3
38	97	103	13882.6	1.2	1.8818e−2	2.1767e−2	109	24	78	6844.75	1.2	9.2783e−3	4.4147e−2

No.	in	out	Len. m	D m	K_m MSm ³ /(h Pa)	K_q MSm ⁶ /(h ² bar ²)	No.	in	out	Len. m	D m	K_m MSm ³ /(h Pa)	K_q MSm ⁶ /(h ² bar ²)
39	103	14	59093.6	1.2	8.0103e−2	5.1135e−3	110	58	119	89179.2	1.2	1.2089e−1	3.3884e−3
40	79	92	74213	1.2	1.0060e−1	4.0717e−3	111	55	43	48033.2	1.2	6.5110e−2	6.2910e−3
41	92	9	10380	1.2	1.4070e−2	2.9111e−2	112	55	47	19465.6	1.2	2.6386e−2	1.5524e−2
42	9	100	39573.1	1.2	5.3642e−2	7.6359e−3	113	47	33	93300.9	1.2	1.2647e−1	3.2387e−3
43	100	15	25443.5	1.2	3.4489e−2	1.1876e−2	114	47	57	31776.6	1.2	4.3074e−2	9.5094e−3
44	92	13	81583.8	1.2	1.1059e−1	3.7039e−3	115	57	111	75196.4	1.2	1.0193e−1	4.0185e−3
45	13	68	49535.2	1.2	6.7146e−2	6.1002e−3	116	57	135	61179.7	1.2	8.2931e−2	4.9392e−3
46	7	65	60573.5	1.2	8.2109e−2	4.9886e−3	117	57	135	61179.7	1.2	8.2931e−2	4.9392e−3
47	12	27	19740	1.2	2.6758e−2	1.5308e−2	118	135	112	27318.1	1.2	3.7031e−2	1.1061e−2
48	10	12	25479.1	1.2	3.4538e−2	1.1860e−2	119	135	113	26466.9	1.2	3.5877e−2	1.1417e−2
49	65	66	94839.2	1.2	1.2856e−1	3.1862e−3	120	18	24	22659.1	1.2	3.0715e−2	1.3336e−2
50	66	30	89395.5	1.2	1.2118e−1	3.3802e−3	121	114	62	37046	1.2	5.0217e−2	8.1568e−3
51	30	84	67770.9	1.2	9.1865e−2	4.4588e−3	122	115	62	37887.8	1.2	5.1358e−2	7.9756e−3
52	84	17	10478.4	1.2	1.4204e−2	2.8838e−2	123	62	54	51079.8	1.2	6.9240e−2	5.9158e−3
53	17	8	9209.88	1.2	1.2484e−2	3.2810e−2	124	62	106	67517.7	1.2	9.1522e−2	4.4755e−3
54	28	51	35640.1	1.2	4.8311e−2	8.4786e−3	125	77	4	3000.01	1.2	4.0666e−3	1.0073e−1
55	17	37	33603.3	1.2	4.5550e−2	8.9925e−3	126	33	74	28202.2	1.2	3.8229e−2	1.0715e−2
56	72	21	96065	1.2	1.3022e−1	3.1455e−3	127	74	23	56443.9	1.2	7.6511e−2	5.3536e−3
57	72	25	61286.2	1.2	8.3075e−2	4.9306e−3	128	23	41	78489.8	1.2	1.0640e−1	3.8499e−3
58	72	22	60740.5	1.2	8.2335e−2	4.9749e−3	129	74	36	50533.9	1.2	6.8500e−2	5.9797e−3
59	129	75	32476.9	1.2	4.4023e−2	9.3043e−3	130	69	36	12358.4	1.2	1.6752e−2	2.4451e−2
60	130	20	26313.7	1.2	3.5669e−2	1.1484e−2	131	26	71	52638.3	1.2	7.1353e−2	5.7406e−3
61	75	76	3274.84	1.2	4.4391e−3	9.2272e−2	132	74	44	22865.5	1.2	3.0995e−2	1.3215e−2
62	76	49	54491	1.2	7.3864e−2	5.5454e−3	133	44	27	70219.5	1.2	9.5185e−2	4.3033e−3
63	49	50	17417.5	1.2	2.3610e−2	1.7349e−2	134	54	29	6799.79	1.2	9.2173e−3	4.4439e−2
64	50	67	77016.1	1.2	1.0440e−1	3.9236e−3	135	29	27	60895.6	1.2	8.2546e−2	4.9622e−3
65	51	38	12536.2	1.2	1.6993e−2	2.4104e−2	136	54	95	73579.3	1.2	9.9739e−2	4.1068e−3
66	67	16	9158.22	1.2	1.2414e−2	3.2995e−2	137	95	61	7053.13	1.2	9.5607e−3	4.2843e−2
67	67	16	9158.22	1.2	1.2414e−2	3.2995e−2	138	27	61	44719	1.2	6.0618e−2	6.7572e−3
68	67	52	8869.54	1.2	1.2023e−2	3.4069e−2	139	61	19	38054.1	1.2	5.1583e−2	7.9407e−3
69	67	52	8869.54	1.2	1.2023e−2	3.4069e−2	140	61	1	15990.8	1.2	2.1676e−2	1.8897e−2
70	20	39	86829.9	1.2	1.1770e−1	3.4801e−3	141	81	48	9159.01	1.2	1.2415e−2	3.2992e−2
71	128	60	85824.9	1.2	1.1634e−1	3.5208e−3							

7.3 Power-Side Units

This subsection lists all power-side generation units, including the reference-bus generator and the GT units coupled to the gas network. The Status column uses 1 for online units and 0 for out-of-service units.

Table 19: P200_G135 Power-Side Generation Units.

No.	Power bus	Type	Gas node	Eff. p.u.	$V_{g,\min}$ p.u.	$V_{g,\max}$ p.u.	$P_{\min}$ MW	$P_{\max}$ MW	$Q_{\min}$ MVA _r	$Q_{\max}$ MVA _r	Status
1	49	GT	15	0.48	0.90	1.10	1.36	4.53	-0.55	2.11	1
2	50	GT	36	0.46	0.90	1.10	1.36	4.53	-0.55	2.11	1
3	51	GT	69	0.45	0.90	1.10	1.36	4.53	-0.55	2.11	1
4	52	GT	131	0.55	0.90	1.10	1.36	4.53	-0.55	2.11	1
5	53	GT	132	0.52	0.90	1.10	2.72	9.07	-1.11	4.23	1
6	65	GP	—	—	0.90	1.10	86.50	86.50	-21.66	32.04	1
7	67	GT	134	0.50	0.90	1.10	1.41	4.70	-0.57	2.19	1
8	68	GP	—	—	0.90	1.10	8.38	27.92	-3.41	13.01	1
9	69	GP	—	—	0.90	1.10	8.38	27.92	-3.41	13.01	1
10	70	GP	—	—	0.90	1.10	8.38	27.92	-3.41	13.01	1
11	71	GP	—	—	0.90	1.10	8.38	27.92	-3.41	13.01	1
12	72	GP	—	—	0.90	1.10	8.38	27.92	-3.41	13.01	1
13	73	GP	—	—	0.90	1.10	8.38	27.92	-3.41	13.01	1
14	76	GP	—	—	0.90	1.10	1.20	4.00	-0.44	2.04	1
15	77	GP	—	—	0.90	1.10	0.72	2.40	-0.27	1.22	1
16	78	GP	—	—	0.90	1.10	5.40	18.00	-2.00	9.16	0
17	79	GP	—	—	0.90	1.10	5.40	18.00	-2.00	9.16	0
18	90	GP	—	—	0.90	1.10	0.96	3.20	-0.36	1.63	1
19	91	GP	—	—	0.90	1.10	1.50	5.00	-0.55	2.55	1
20	92	GP	—	—	0.90	1.10	1.89	6.30	-0.70	3.21	0
21	94	GP	—	—	0.90	1.10	5.40	18.00	-2.20	8.39	1
22	104	GP	—	—	0.90	1.10	67.60	67.60	-14.26	21.09	1
23	105	GP	—	—	0.90	1.10	154.80	154.80	-28.51	42.17	1
24	114	GP	—	—	0.90	1.10	1.40	1.40	-0.24	0.36	1
25	115	GP	—	—	0.90	1.10	133.50	133.50	-21.60	31.95	1
26	125	GP	—	—	0.90	1.10	39.02	130.05	-15.87	60.60	1
27	126	GP	—	—	0.90	1.10	39.02	130.05	-15.87	60.60	1
28	127	GP	—	—	0.90	1.10	39.02	130.05	-15.87	60.60	1
29	135	GP	—	—	0.90	1.10	133.92	446.40	-54.46	208.02	1
30	136	GP	—	—	0.90	1.10	133.92	446.40	-54.46	208.02	1
31	147	GP	—	—	0.90	1.10	92.40	92.40	-14.47	21.41	1
32	151	GP	—	—	0.90	1.10	1.62	5.40	-0.66	2.52	1
33	152	GP	—	—	0.90	1.10	23.17	77.22	-9.42	35.99	1
34	153	GP	—	—	0.90	1.10	23.17	77.22	-9.42	35.99	1
35	154	GP	—	—	0.90	1.10	23.17	77.22	-9.42	35.99	1
36	155	GP	—	—	0.90	1.10	23.17	77.22	-9.42	35.99	1
37	161	GP	—	—	0.90	1.10	41.58	138.60	-15.38	70.55	0
38	164	GP	—	—	0.90	1.10	3.60	12.00	-1.33	6.11	0

No.	Power bus	Type	Gas node	Eff. p.u.	$V_{g,\min}$ p.u.	$V_{g,\max}$ p.u.	$P_{\min}$ MW	$P_{\max}$ MW	$Q_{\min}$ MVA	$Q_{\max}$ MVA	Status
39	165	GP	—	—	0.90	1.10	7.80	26.00	-2.89	13.23	0
40	166	GP	—	—	0.90	1.10	2.82	9.40	-1.04	4.78	0
41	167	GP	—	—	0.90	1.10	2.82	9.40	-1.04	4.78	1
42	168	GP	—	—	0.90	1.10	2.82	9.40	-1.04	4.78	0
43	169	GP	—	—	0.90	1.10	2.82	9.40	-1.04	4.78	0
44	170	GP	—	—	0.90	1.10	2.82	9.40	-1.04	4.78	1
45	182	GP	—	—	0.90	1.10	5.25	17.50	-2.14	8.16	1
46	183	GP	—	—	0.90	1.10	7.98	26.60	-3.25	12.40	1
47	189	GP	—	—	0.90	1.10	170.75	569.15	-46.67	209.45	1
48	196	GP	—	—	0.90	1.10	20.25	67.50	-7.49	34.36	0
49	197	GP	—	—	0.90	1.10	20.25	67.50	-7.49	34.36	0

7.4 Electric-Gas Coupling Elements

This subsection reports P2G units and compressors that couple the electric and gas subsystems.

Table 20: P200_G135 P2G Units.

No.	Power bus	Gas node	Eff. p.u.	$P_{\min}$ MW	$P_{\max}$ MW	Coupling strength
1	135	131	0.45	0.00	10.00	60.00
2	136	132	0.45	0.00	10.00	60.00
3	127	134	0.45	0.00	10.00	60.00
4	125	15	0.45	0.00	10.00	60.00
5	126	36	0.45	0.00	50.00	60.00
6	152	69	0.45	0.00	30.00	60.00
7	153	82	0.45	0.00	10.00	60.00
8	154	96	0.45	0.00	10.00	60.00
9	155	105	0.45	0.00	10.00	60.00

Table 21: P200_G135 Compressor Units; E and G denote electric-driven and gas-driven compressors, respectively.

No.	In.	Out.	Drv.	Power bus	Ratio p.u.	Power MW	No.	In.	Out.	Drv.	Power bus	Ratio p.u.	Power MW
1	18	131	E	135	1.00–1.05	0–60	16	58	134	E	127	1.00–1.05	0–10
2	77	106	G	136	1.00–1.05	—	17	57	118	G	142	1.00–1.05	—
3	67	107	G	135	1.00–1.05	—	18	57	119	G	148	1.00–1.05	—
4	67	108	E	136	1.00–1.05	0–10	19	45	120	E	135	1.00–1.05	0–10
5	64	109	G	125	1.00–1.05	—	20	45	121	G	136	1.00–1.05	—
6	64	110	G	126	1.00–1.05	—	21	45	122	G	135	1.00–1.05	—
7	35	132	E	127	1.00–1.05	0–10	22	45	123	E	136	1.00–1.05	0–10

No.	In.	Out.	Drv.	Power bus	Ratio p.u.	Power MW	No.	In.	Out.	Drv.	Power bus	Ratio p.u.	Power MW
8	35	133	G	142	1.00–1.05	—	23	31	124	G	125	1.00–1.05	—
9	6	111	G	148	1.00–1.05	—	24	31	125	G	126	1.00–1.05	—
10	6	112	E	135	1.00–1.05	0–40	25	60	126	E	127	1.00–1.05	0–60
11	6	113	G	136	1.00–1.05	—	26	60	127	G	142	1.00–1.05	—
12	114	135	G	135	1.00–1.05	—	27	50	128	G	148	1.00–1.05	—
13	115	135	E	136	1.00–1.05	0–10	28	72	129	E	135	1.00–1.05	0–80
14	58	116	G	125	1.00–1.05	—	29	72	130	G	136	1.00–1.05	—
15	58	117	G	126	1.00–1.05	—							

8 Reproducibility Note

The benchmark descriptions in this companion document are intended to be read as a deterministic data specification for the three IEGS cases. The tabulated gas wells, gas-network nodes, ordinary pipelines, power-side generation units, P2G units, and compressors are the effective benchmark data used by the corresponding integrated electric–gas OEF and reinforcement-learning studies. The same notation, units, and coupling conventions are used across P14_G11, P57_G40, and P200_G135, so that differences among the three cases reflect benchmark scale and topology rather than changes in data interpretation.

The companion source code for the integrated electric–gas system benchmarks is openly available at <https://github.com/bbYCH/Article-Companion-for-The-Topology-of-Integrated-Electric-Gas-Systems>.